Discrete Almgren-Chriss Optimal Execution

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1 Model Setup and Market Dynamics

Definition 1.1 (Trading Schedule)**.** *Let T be the total time horizon and $N \in \mathbb{N}$ the number of discrete trading intervals. We define the time step $\tau = \frac{T}{N}$ and the discrete time grid $t_k = k\tau$ for $k \in \{0, 1, \dots, N\}$.*

Let X denote the initial inventory. The inventory at time t_k is denoted x_k, subject to the boundary conditions $x_0 = X$ and $x_N = 0$. The number of shares liquidated between t_{k-1} and t_k is given by:

$$v_k = x_{k-1} - x_k \quad \textit{for} \quad k \in \{1, \dots, N\}$$

Definition 1.2 (Price Dynamics and Execution Cost)**.** *The fundamental price process follows an arithmetic random walk with permanent market impact:*

$$S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma v_k$$

where $\xi_k \overset{iid}{\sim} \mathcal{N}(0,1)$, σ is the arithmetic volatility, and $\gamma \geq 0$ is the permanent impact parameter.

The actual execution price incorporates a temporary market impact penalty $\eta \geq 0$:

$$\tilde{S}_k = S_{k-1} - \eta\frac{v_k}{\tau}$$

The total cost of execution is the sum of cash inflows:

$$C(x) = \sum_{k=1}^{N} v_k \tilde{S}_k$$

2 Implementation Shortfall and Objective Function

Definition 2.1 (Implementation Shortfall)**.** *The Implementation Shortfall (IS) measures the execution cost relative to the initial paper value of the portfolio:*

$$IS = C(x) - XS_0$$

Proposition 2.2 (Moments of the Implementation Shortfall)**.** *The expected value and variance of the Implementation Shortfall are given by:*

$$E[IS] = -\sum_{k=1}^{N}\left(v_k\gamma(x_0 - x_{k-1}) + \eta\frac{v_k^2}{\tau}\right)$$
$$V[IS] = \sigma^2\tau\sum_{i=1}^{N-1} x_i^2$$

Proof. The Implementation Shortfall is given by:

$$IS = \sum_{k=1}^{N} \left(v_k S_{k-1} - \eta \frac{v_k^2}{\tau} \right) - XS_0$$

Substituting the unrolled fundamental price $S_{k-1} = S_0 + \sigma\sqrt{\tau}\sum_{i=1}^{k-1}\xi_i - \gamma(x_0 - x_{k-1})$, the expected value extracts the deterministic components, yielding $E[IS]$.

For the variance, we isolate the stochastic term of IS:

$$IS_{\text{stochastic}} = \sigma\sqrt{\tau}\sum_{k=1}^{N} v_k \sum_{i=1}^{k-1} \xi_i$$

By swapping the order of summation over the discrete indices from $\{(k,i) : 1 \le k \le N, 1 \le i \le k-1\}$ to $\{(i,k) : 1 \le i \le N-1, i+1 \le k \le N\}$, we obtain:

$$IS_{\text{stochastic}} = \sigma\sqrt{\tau}\sum_{i=1}^{N-1} \xi_i \sum_{k=i+1}^{N} v_k$$

Since the sum of executed volumes from $i+1$ to N is exactly the remaining inventory $x_i - x_N$, and $x_N = 0$, this simplifies to $\sigma\sqrt{\tau}\sum_{i=1}^{N-1} x_i\xi_i$. Because ξ_i are independent standard normal variables, the variance evaluates to the sum of the squared coefficients:

$$V[IS] = \sigma^2\tau\sum_{i=1}^{N-1} x_i^2$$

□

Definition 2.3 (Mean-Variance Objective Function)**.** *Because IS represents a negative loss, we aim to minimize the expected execution penalty $E = -E[IS]$ penalized by the strategy's variance. For a risk aversion parameter $\lambda \ge 0$, the objective function is:*

$$U(x) = E + \lambda V[IS]$$

Proposition 2.4 (Quadratic Form of the Objective Function)**.** *The objective function can be isolated with respect to the interior points $x = (x_1, x_2, \ldots, x_{N-1})$ as a strict multi-dimensional quadratic form $\tilde{U}(x) = \frac{1}{2}x^T Ax - b^T x$, where $\tilde{U}(x)$ is defined as:*

$$\tilde{U}(x) = \frac{\tilde{\eta}}{\tau}\sum_{k=1}^{N}(x_{k-1} - x_k)^2 + \lambda\sigma^2\tau\sum_{k=1}^{N-1} x_k^2 \qquad \text{with } \tilde{\eta} = \eta - \frac{1}{2}\gamma\tau$$

The elements of the symmetric tridiagonal matrix A and vector b are:

$$A_{k,k} = \frac{4\tilde{\eta}}{\tau} + 2\lambda\sigma^2\tau$$
$$A_{k,k+1} = A_{k+1,k} = -\frac{2\tilde{\eta}}{\tau}$$
$$b_1 = \frac{2\tilde{\eta}X}{\tau}, \quad b_k = 0 \quad \text{for } k > 1$$

Proof. We expand the impact penalty sum $\sum_{k=1}^{N}(x_{k-1} - x_k)^2$:

$$\sum_{k=1}^{N}(x_{k-1} - x_k)^2 = x_0^2 - 2x_0x_1 + 2\sum_{k=1}^{N-1} x_k^2 - 2\sum_{k=1}^{N-2} x_kx_{k+1} + x_N^2$$

Since $x_0 = X$ and $x_N = 0$, substituting this into $\tilde{U}(x)$ yields:

$$\tilde{U}(x) = \frac{\tilde{\eta}}{\tau}X^2 - \frac{2\tilde{\eta}}{\tau}Xx_1 - \frac{2\tilde{\eta}}{\tau}\sum_{k=1}^{N-2} x_k x_{k+1} + \sum_{k=1}^{N-1}\left(\frac{2\tilde{\eta}}{\tau} + \lambda\sigma^2\tau\right)x_k^2$$

By mapping this to the standard quadratic form $\frac{1}{2}x^TAx - b^Tx$, we multiply the x_k^2 coefficients by 2 to identify the diagonal elements $A_{k,k} = \frac{4\tilde{\eta}}{\tau} + 2\lambda\sigma^2\tau$, identify the off-diagonal elements $A_{k,k+1} = -\frac{2\tilde{\eta}}{\tau}$, and extract the linear vector component $b_1 = \frac{2\tilde{\eta}X}{\tau}$. □

Theorem 2.5 (Strict Convexity and Global Minimum)**.** *The matrix $A = \nabla^2\tilde{U}(x)$ is positive definite, meaning for all non-zero vectors $x \in \mathbb{R}^{N-1}$, $x^TAx > 0$. Consequently, $\tilde{U}(x)$ is strictly convex and admits a unique global minimum x^*.*

Proof. For any non-zero vector $x \in \mathbb{R}^{N-1}$, the quadratic form expands as:

$$x^TAx = \left(\frac{4\tilde{\eta}}{\tau} + 2\lambda\sigma^2\tau\right)\sum_{k=1}^{N-1} x_k^2 - \frac{4\tilde{\eta}}{\tau}\sum_{k=1}^{N-2} x_k x_{k+1}$$

Factoring out $\frac{2\tilde{\eta}}{\tau}$, we rewrite the sum:

$$x^TAx = \frac{2\tilde{\eta}}{\tau}\left(2\sum_{k=1}^{N-1} x_k^2 - 2\sum_{k=1}^{N-2} x_k x_{k+1}\right) + 2\lambda\sigma^2\tau\sum_{k=1}^{N-1} x_k^2$$

Applying the algebraic identity $(a-b)^2 = a^2 - 2ab + b^2$ allows us to regroup the bracketed terms:

$$x^TAx = \frac{2\tilde{\eta}}{\tau}\left[x_1^2 + x_{N-1}^2 + \sum_{k=1}^{N-2}(x_k - x_{k+1})^2\right] + 2\lambda\sigma^2\tau\sum_{k=1}^{N-1} x_k^2$$

Since $\tilde{\eta}, \tau, \lambda, \sigma^2 > 0$, every term is a non-negative sum of squares. For $x \neq 0$, at least one coordinate is non-zero, ensuring $\sum x_k^2 > 0$. Thus, $x^TAx > 0$. □

Proposition 2.6 (Risk-Neutral Execution Limit (TWAP))**.** *As the risk aversion parameter $\lambda \to 0$, the optimal execution strategy converges to the Time-Weighted Average Price (TWAP) schedule, where the executed volume at each step is strictly constant: $v_k = \frac{X}{N}$ for all $k \in \{1, \ldots, N\}$.*

Proof. For a risk-neutral agent ($\lambda = 0$), the objective function reduces entirely to the temporary market impact penalty:

$$\min_{\mathbf{v}} \sum_{k=1}^{N} v_k^2$$

subject to the liquidation constraint $\sum_{k=1}^{N} v_k = X$.

Let $f(v) = v^2$. Since the second derivative $f''(v) = 2 > 0$, the function is strictly convex. Applying Jensen's inequality with uniform weights $w_k = \frac{1}{N}$:

$$f\left(\frac{1}{N}\sum_{k=1}^{N} v_k\right) \leq \frac{1}{N}\sum_{k=1}^{N} f(v_k)$$

Substituting the constraint $\sum_{k=1}^{N} v_k = X$ into the left side yields:

$$\left(\frac{X}{N}\right)^2 \leq \frac{1}{N}\sum_{k=1}^{N} v_k^2 \implies \sum_{k=1}^{N} v_k^2 \geq \frac{X^2}{N}$$

Because $f(v)$ is strictly convex, Jensen's inequality holds as a strict equality if and only if all variables are identical. Therefore, the absolute minimum is uniquely achieved when $v_1 = v_2 = \cdots = v_N = \frac{X}{N}$. □

Remark 2.7 (High Risk Aversion Boundary)**.** *Conversely, as λ increases, the variance penalty $\sum_{k=1}^{N-1} x_k^2$ dominates the objective function. To minimize this holding area, the optimization algorithm generates a highly convex, front-loaded schedule, aggressively liquidating inventory in the earliest trading intervals to shed stochastic risk.*

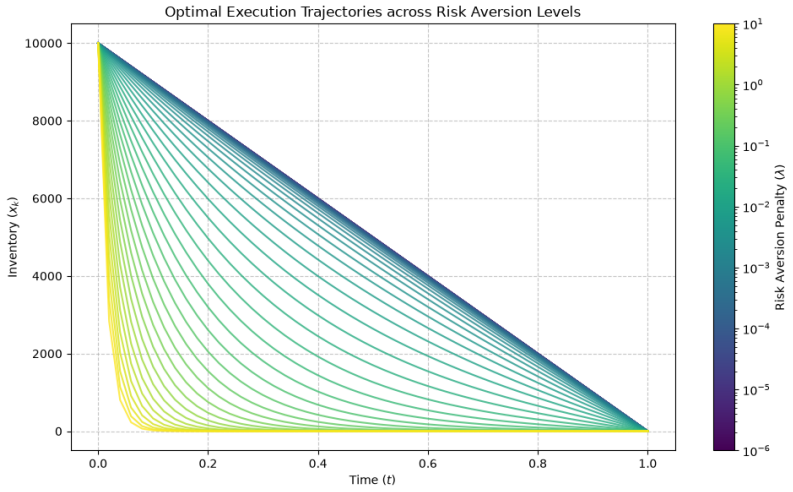


Figure 1: We implemented the Almgren-Chriss model for 50 different values of λ ranging from 10^{-6} to 10. We can clearly observe how the curvature evolves as λ gets bigger and how it tends to become a straight line as $\lambda -> 0$

3 Continuous-Time Limit and ODE Formulation

Theorem 3.1 (Continuous Limit of the Difference Equation)**.** *The first-order condition $\nabla \tilde{U}(x^*) = 0 \implies Ax^* = b$ implies that for the interior points $1 < k < N-1$, the sequence satisfies:*

$$\frac{x_{k+1} - 2x_k + x_{k-1}}{\tau^2} = \frac{\lambda \sigma^2}{\tilde{\eta}} x_k$$

Taking the limit as $\tau \to 0$, this difference equation converges to the ordinary differential equation:

$$x''(t) - \tilde{\kappa}^2 x(t) = 0$$

where $\tilde{\kappa} = \sigma \sqrt{\frac{\lambda}{\tilde{\eta}}}$.

Proof. Extracting the k-th row from the linear system $Ax = b$ for an interior point gives:

$$-\frac{2\tilde{\eta}}{\tau} x_{k-1} + \left(\frac{4\tilde{\eta}}{\tau} + 2\lambda \sigma^2 \tau \right) x_k - \frac{2\tilde{\eta}}{\tau} x_{k+1} = 0$$

Dividing by $-\frac{2\tilde{\eta}}{\tau}$ and rearranging the terms yields:

$$x_{k+1} - 2x_k + x_{k-1} = \frac{\lambda \sigma^2 \tau^2}{\tilde{\eta}} x_k$$

Dividing both sides by τ^2 produces the discrete second derivative operator on the left side. As $\tau \to 0$, the left side formally converges to $x''(t)$, yielding $x''(t) - \tilde{\kappa}^2 x(t) = 0$. □

Proposition 3.2 (Exact Analytical Solution)**.** *The continuous-time ODE subject to the boundary conditions $x(0) = X$ and $x(T) = 0$ admits the closed-form analytical solution:*

$$x(t) = X \frac{\sinh(\tilde{\kappa}(T-t))}{\sinh(\tilde{\kappa} T)}$$

Proof. The characteristic equation for the ODE is $r^2 - \tilde{\kappa}^2 = 0$, yielding roots $r = \pm\tilde{\kappa}$. The general solution in exponential basis is:

$$x(t) = Ae^{\tilde{\kappa}t} + Be^{-\tilde{\kappa}t}$$

Applying the boundary conditions $x(0) = X$ and $x(T) = 0$ yields a system of equations for the constants, resolving to:

$$A = \frac{-e^{-\tilde{\kappa}T}}{2\sinh(\tilde{\kappa}T)}X, \quad B = \frac{e^{\tilde{\kappa}T}}{2\sinh(\tilde{\kappa}T)}X$$

Substituting these constants back into the general solution:

$$x(t) = \frac{X}{2\sinh(\tilde{\kappa}T)}\left(e^{\tilde{\kappa}(T-t)} - e^{-\tilde{\kappa}(T-t)}\right)$$

By the definition of the hyperbolic sine, $\sinh(z) = \frac{e^z - e^{-z}}{2}$, this simplifies strictly to:

$$x(t) = X\frac{\sinh(\tilde{\kappa}(T-t))}{\sinh(\tilde{\kappa}T)}$$

□

Remark 3.3. *The continuous model confirms that the discrete matrix mechanics align strictly with the market dynamics. The structural constant $\tilde{\kappa}$ governs the curvature of the liquidation scheme; a higher $\tilde{\kappa}$ implies higher execution urgency. Numerically, evaluating the continuous solution against the discrete $O(N)$ solver yields deviations on the order of 0.11%, isolated entirely to discretization error.*

4 Empirical Validation and Efficient Frontier Mechanics

To rigorously validate the discrete mathematical filtrations of the objective function, an event-driven Monte Carlo engine was constructed to simulate $M = 10,000$ price paths.

Implied Parameter Calibration Due to the absence of proprietary limit order book data, the market impact parameters were determined via implied calibration. The parameters were algebraically reverse-engineered to ensure the risk-neutral TWAP strategy yields a standard institutional friction cost of roughly 25 basis points (bps) for a $\$1,000,000$ mid-cap equity position ($X = 10,000$ shares, $S_0 = 100$).

Setting the temporary impact $\eta = 2 \times 10^{-5}$ and permanent impact $\gamma = 1 \times 10^{-5}$ over $N = 50$ intervals ($\tau = 0.02$) anchors the baseline TWAP execution cost to exactly $\$2,490$ (24.9 bps). The arithmetic volatility is set to $\sigma = 0.3$. This scales the maximum time-exposure risk to approximately 17 bps, precisely reflecting the realistic standard deviation of an intraday execution.

Proposition 4.1 (Maximum Variance Limit)**.** *The physical boundary of the Efficient Frontier is strictly bounded on the right by the risk-neutral TWAP strategy ($\lambda \to 0$). The absolute maximum variance exposure is given by:*

$$V_{max} = \sigma^2\tau\sum_{j=1}^{N-1}\left(X\frac{j}{N}\right)^2$$

Proof. By substituting the deterministic TWAP inventory schedule $x_k = X\frac{N-k}{N}$ into the variance equation $V[IS] = \sigma^2\tau\sum_{k=1}^{N-1}x_k^2$, we obtain:

$$V_{max} = \sigma^2\tau\sum_{k=1}^{N-1}\left(X\frac{N-k}{N}\right)^2$$

Changing the summation index to $j = N - k$ reverses the sum but preserves the terms precisely, yielding the result. The variance cannot physically exceed this value, as TWAP represents the maximum possible time-exposure of the inventory to stochastic price shocks. With our calibrated parameters, $V_{max} \approx 2.91 \times 10^6$, corresponding to an execution standard deviation of $\$1,706$ (17 bps). □

Structural Bounds vs. The Institutional Operating Window To understand the global properties of the continuous ODE versus the local constraints of a trading desk, we isolate the Efficient Frontier into two distinct scopes.

Figure 2 displays the macro-mathematical view ($\lambda \in [10^{-6}, 10^{1}]$). As $\lambda \to \infty$, the variance penalty forces a corner solution: an instantaneous block liquidation ($v_1 \to X$). Under the infinite liquidity assumption of the linear framework, this yields a theoretical shortfall of approximately $-\$560,000$. In practice, this 56% slippage represents a physical liquidity wall—a mathematically valid limit that no rational portfolio manager would ever execute.

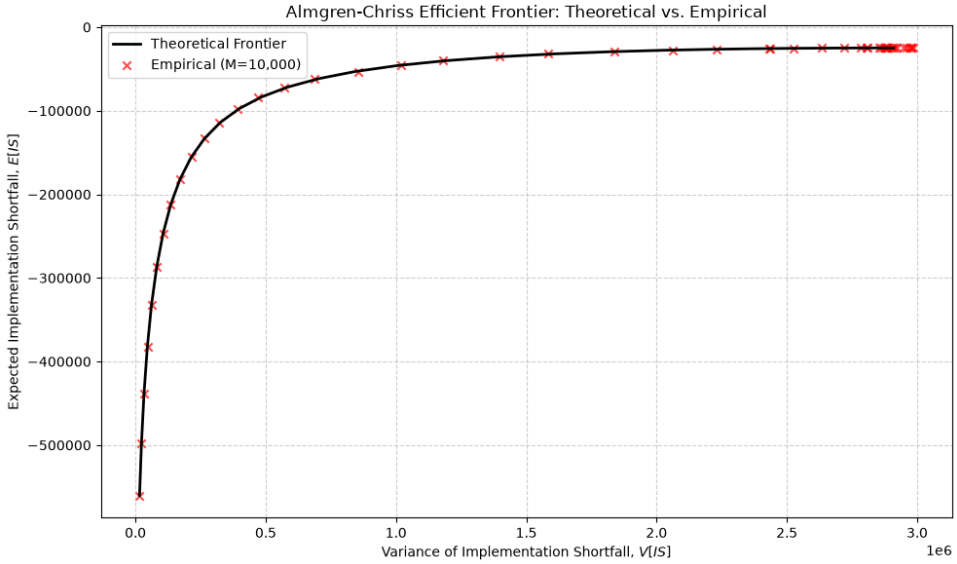


Figure 2: Theoretical Bounds ($\lambda \in [10^{-6}, 10^{1}]$). The left boundary represents the prohibitive market impact of an instantaneous block liquidation.

Figure 3 restricts the domain to the viable operational window ($\lambda \in [10^{-6}, 10^{-1}]$). Here, execution costs range from the 24.9 bps TWAP baseline up to approximately 200 bps. Quantitative funds restrict their optimization entirely to this flatter sub-domain. The exact chosen coordinate balances minor temporary impact penalties against strategy-specific constraints, such as alpha decay (directional drift), preserving tight Sharpe ratios in statistical arbitrage, or preempting volatility shocks.

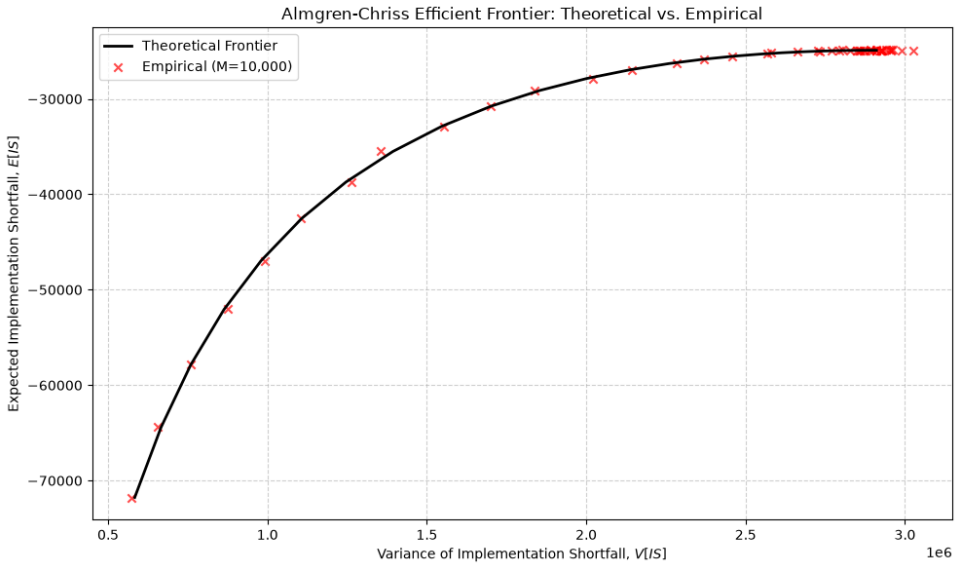


Figure 3: The Rational Execution Window ($\lambda \in [10^{-6}, 10^{-1}]$). The strict operational domain where execution algorithms balance impact costs against standard intraday variance.

Invariance to Permanent Impact While the permanent impact parameter γ explicitly penalizes the total expected shortfall (shifting the Efficient Frontier vertically along the y-axis), it entirely vanishes from the Hessian matrix A and the linear vector b in the quadratic form $\tilde{U}(x)$. Mathematically, γ plays absolutely no role in shaping the optimal execution trajectory x^*.